[INCH-POUND]
GGG-S-656E
January 29, 1996
SUPERSEDING
GGG-S-656D
September 25, 1987

FEDERAL SPECIFICATION

SQUARES, CARPENTERS', DIEMAKERS', AND MACHINISTS'

The General Services Administration has authorized the use of this Federal Specification by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers several types, classes, and styles of squares (with or without attachments).

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. The squares covered by this specification shall be of the following types, classes, and styles as specified (see 6.1):

Type I - Carpenter's, steel, combination try and miter
Type II - Carpenters', one piece
  Class A - Steel
    Style 1 - Tapered
    Style 2 - Parallel
Type III - Carpenters', try
    Style 1 - Try
    Style 2 - Try and miter
Type IV - Diemakers', with interchangeable blades
Type V - Diemakers', with interchangeable blades adjustable at an angle
Type VI - Machinists' combination (for center head and protractor head, see 3.18.1)
  Class A - Square and miter head; cast iron
Type VII - Machinists', double square, with interchangeable blades
Type VIII - Machinists', thin steel
Type IX - Machinists', try

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any other data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406.

FSC 5210

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the current version before use.

Federal Specification:

PPP-P-40 - Preservation and Packing of Hand Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery

Federal Standard:

FED-STD-H28 - Screw - Thread Standards for Federal Services

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification, and other Federal specifications and Commercial Item Descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI)/American Society of Quality Control (ASQC):

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American Society for Quality Control, P. O. Box 3005, 611 E. Wisconsin Avenue, Milwaukee, WI 53201-4606.)

American Society for Testing and Materials (ASTM):

ASTM E 18 - Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of squares which are otherwise in accordance with this specification.

3.2 Material. The material shall be as hereinafter specified for each type and class.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the currentversion before use.

3.2.1 Marking. Each item shall be marked with the manufacturer's name or identifying symbol and the state or country of manufacture. The marking shall be engraved, etched, or stamped in such a manner that it will be permanent to the extent that it will remain clear and legible throughout the life of the item.

3.3 Design. The design of each square shall be similar to the applicable figure and shall conform to the quality and dimensions hereinafter specified for each type, class, and style. Unless otherwise specified, standard commercial tolerances shall apply to all dimensions.

3.4 Screw threads. Screw thread construction and tolerances shall conform to FED-STD-H28.

3.5 Finish and coatings. All utile or machine-ground and polished surfaces including surfaces bearing graduations, except for types I, II, III, and IX shall have a finish of 32 microinches roughness height rating or better. Types I, II, and III squares shall have a finish of 100 microinches roughness height rating or better. Type IX squares shall have a finish of 16 microinches roughness height rating or better. Determination of surface finish of blade edges shall be made by use of standard surface finish masters for visual comparison. Graduations and figures shall have a finish which will make it possible for each accurate reading. Graduations and figures of types I, II, and III squares shall have a durable filling of white, yellow, black, or other contrasting color. Component parts which are fabricated of forged steel or cast iron shall have a durable dark colored, smooth, or rippled enamel finish on surfaces that are not machined or polished.

3.6 Graduations. All graduation lines shall be etched or machine cut to a width and depth as hereinafter specified for each type and class. Types I and III squares may have etched, machine cut, or stamped graduations. The edges of graduation lines shall be sharp, straight, legible, and permanent. Graduation lines representing equal fractional aliquots of any scale, in descending order, shall be of equal height and readily distinguishable by their relative height from graduation lines representing other fractional aliquots, except that on scales graduated to 50th and 100ths of an inch, alternate graduation lines or alternate pairs of lines may be of equal height, or of such individual heights that the minimum graduations are easily readable. The maximum variation in width between like types of graduations and in the individual graduation shall be such that the maximum width shall not exceed the minimum width by more than 35 percent of the minimum width. Graduations shall extend to the reading edge.

3.7 Scale accuracy of graduated blades. The scale accuracy of the graduated blades shall be as hereinafter specified.

3.7.1 Blade length. When maximum and minimum length is not specified, blade length shall be at least the length shown on applicable tables.

3.8 Figures. Each figure shall be sunk below the plane of the face on which it appears. Figures shall be clear, distinct, and permanent.

3.9 Level glasses.

3.9.1 Construction. Level glasses shall be fabricated of substantial glass or plastic tubing properly formed with permanent and readily distinguishable black bubble defining lines, suitable for properly centering the bubble. Level glasses shall be permanently secured to the head and enclosed by a metal housing in a manner such that only the portion necessary for observing the bubble positions is exposed. The liquid in the glass level shall be either clear or amber colored ("Cats Eye"), at the option of the manufacturer. The liquid shall not discolor, cloud, or noticeably change in fluidity with age or at temperatures as low as - 20° Farenheit (F) and as high as 125°F.

3.9.2 Vial accuracy. The level vial shall have a vial sensitivity not to exceed 50 minutes and shall be set in the square head parallel to the base within 0.012 inch per foot.

3.10 Interchangeability. Individual movable parts of identical models of squares made by the same manufacturer shall be readily interchangeable and within the accuracy's specified without the necessity of any alteration, refitting, or adjustment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

3.11 Heat treatment. Type VI cast iron square and miter heads, center and protractor heads, and type VII cast iron heads shall be stress relieved. Types IV, V, and IX beams shall have a hardness of not less than 50 on the Rockwell C scale. Unless otherwise specified, all other hardened parts shall have a minimum hardness of 40 on the Rockwell C scale.

3.12 Accuracy of squares. The square and miter accuracy shall conform to that specified herein for the type, class, style, and size furnished (see 6.1).

3.13 Type I, carpenters', steel, combination try and miter. Type I squares shall be similar to Figure 1 and shall conform to the dimensions specified in Table I for the length of blade specified (see 6.1).

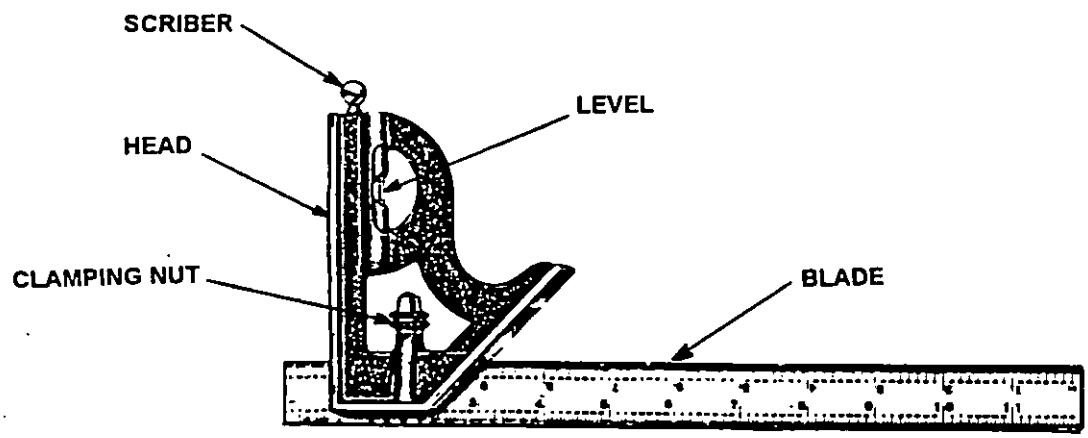


Figure 1. Type I, carpenters', steel, combination try and miter.

TABLE I. Type I, carpenters', steel, combination try and miter

Length of blade (Inches)	Width of blade		Thickness of blade		Length of head	
	Minimum (Inch)	Maximum (Inches)	Minimum (Inch)	Maximum (Inch)	Minimum (Inches)	Maximum (Inches)
9	7/8	1	0.070	0.095	3-1/4	4
12	15/16	1-1/8	0.070	0.095	4	5

3.13.1 Components. The square shall consist of a blade, clampable square and miter head with one or more level glasses, a blade clamping device, and a scriber.

3.13.2 Blade. The blade shall be made of steel, of uniform thickness with a flatness of 0.010 total indicated variation (TIV). The blade shall have a minimum hardness of 90 on the Rockwell B scale and shall be nickel, chrome plated, blued or black oxide. Edges and faces shall be straight and parallel within 0.004 inch per each 12 inch length. A full length groove along the center of one face shall be provided for engagement with the clamping device on the head.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

3.1.3.3 Head. The head shall be made of steel or a fine grained, seasoned gray iron casting of uniform quality, free from blow holes, porosity, hard spots, or other detrimental defects. There shall be a slot for insertion of the blade. The clamping device shall have a substantial knurled thumbscrew or thumbnut suitable for rigidly clamping the head at any point of the blade without play in any direction. The locking device shall be so designed to prevent slipping from the head when the blade has been removed. The level glass shall be parallel with the square face of the head.

3.13.4 Finish. The finish of all parts shall conform to requirements in 3.5.

3.13.5 Accuracy.

3.13.5.1 Graduations and figures. Graduation lines shall be of uniform width and depth. Graduations and figures shall be etched, machine cut, or stamped to a minimum depth of 0.002 inch. Inch lines shall be from 0.008 to 0.022 inch in width, whereas increment lines shall be from 0.008 to 0.020 inch in width. The accumulated error of graduations shall not exceed 0.004 inch in any one inch, or a total of 1/32 inch in 12 inches. Figures on the grooved face shall read consecutively in inches over the total length of the blade with figures of each edge reading in the opposite direction. The blade shall be subdivided in 8ths on one edge and 16ths on the other edge; on the other face of the blade, the subdivisions shall be either 8ths on one edge and 16ths on the other edge, or 16ths on one edge and 32nds on the other edge. The subdivisions shall be marked with the proper figures at the corresponding inches; corners of each graduated edge of each face shall be marked to show the smallest fractional subdivision to which it is graduated.

3.13.5.2 Squareness. The out-of-square error of the head with either edge of the blade shall not exceed 0.001 inch per inch of extended blade when measured at the end of blade (see 4.4.2). The out-of-square error of the face shall not exceed a total of 1/16 inch in 12 inches.

3.13.5.3 Miter. The out-of-miter error of the head with either edge of blade shall not exceed 0.012 inch, measured at either end of the blade.

3.14 Type II, carpenters', one piece. The type II squares shall be similar to Figure 2.

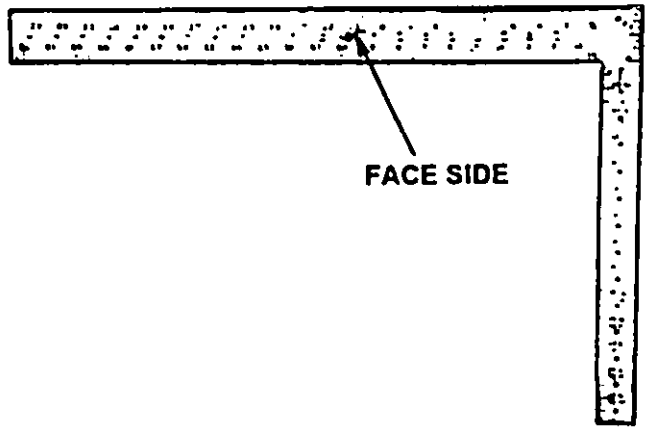


Figure 2. Type II, carpenters', one piece.

3.14.1 Class A, steel. The steel square shall comply with the requirements of Table II for the size number specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE II. Type II, carpenters', one piece

Size	Length of body (beam) along outside edge (Inches)	Length of tongue (blade) along outside edge (Inches)	Width of body (beam) ± 1/32	Width of tongue (blade) ± 1/32	Thickness of tongue (blade) and body (beam) with tapered faces		Thickness of tongue (blade) and body (beam) with parallel faces	
					Ends (Min)	Where tongue (blade) and body (beam) join (Max) (Inch)	(Min)	(Max)
24	24	16	2	1-1/2	0.070	0.135	0.088	0.100
18	18	12	1-1/2	1	---	---	0.088	0.100
12	12	8	1-1/2	1	---	---	0.088	0.100

3.14.1.1 Style 1, tapered. The style 1 tapered square shall have tapered faces on both the tongue (blade) and body (beam) and shall conform to the dimensions of the 24 inch size in Table II.

3.14.1.2 Style 2, parallel. The style 2, parallel squares shall have tongues (blades) and bodies (beams) of equal thickness throughout.

3.14.2 Construction. The square shall be of one-piece construction and shall consist of a body (beam) and tongue (blade). The edges shall be straight and parallel within 0.004 inch per each 12 inch length. The face and back shall be flat.

3.14.3 Finish. Steel squares shall have a bright polished, polished and blued, or equal corrosion-resistant chemical finish.

3.14.4 Graduation lines. Graduation lines shall be of uniform width and depth. Graduations and figures shall be etched, machine cut, or embossed to a minimum depth of 0.0004 inch. Inch and increment lines shall be from 0.008 to 0.020 inch in width. The graduations and figures shall conform to Table III for style group of graduation increments and reference tables.

TABLE III. Style of graduations for type II, carpenters', one piece

Style of graduations	Graduations (inch)				Tables, scales, and measures
	Face		Back		
	Body (beam)	Tongue (blade)	Body (beam)	Tongue (blade)	
A	1/16, 1/8	1/16, 1/8	1/12, 1/16	1/10, 1/12	Brace table Octagon scale Essex board measure 100th scale
B	1/16, 1/4	1/16, 1/4	1/12, 1/4	1/12, 1/4	Brace table Essex board measure
C	1/16, 1/8	1/16, 1/8	1/12, 1/8	1/12, 1/8	None

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE III. Style of graduations for type II, carpenters', one piece (cont'd)

Style of graduations	Graduations (inch)				Tables, scales, and measures
	Face		Back		
	Body (beam)	Tongue (blade)	Body (beam)	Tongue (blade)	
D 1/	1/8, 1/4	1/8, 1/4	1/12, 1/4	1/12, 1/4	None
E	1/8	1/8	1/4	1/4	None

1/ Graduations may be either on the outside edges only or on both inside and outside edges.

3.14.4.1 Size 24. Blade graduations shall be either style A or style B as specified (see 6.1). If the style of graduations is not specified for size 24, style A graduations shall be furnished on class A squares.

3.14.4.2 Size 18. Blade graduations shall be style C.

3.14.4.3 Size 12. Blade graduations shall be style D or E as specified (see 6.1). If the style of graduations is not specified, style D graduations shall be furnished.

3.14.5 Inch marking. Each inch graduation of each edge of both faces (face and back) shall be marked to show the number of inches, except when style E graduations are furnished, in which case each inch graduation on the outer edge of both faces shall be marked to show the number of inches. The size 24, class A square shall have a reverse reading scale in the center of the face of the body unless framing rafter table is required (see 6.1).

3.14.6 Squareness accuracy. The out-of-square error of either edge of the tongue with either edge of the body shall not exceed 0.006 inch per foot, except for size 12 squares with style E graduations, in which case the error shall not exceed 0.015 inch. The out-of-square error shall be determined at the extreme ends of the edges (see 4.4.2).

3.14.7 Scale accuracy. The accumulated error of graduations shall not exceed 1/32 inch per 12 inches.

3.14.8 Instructions. A booklet shall be furnished with each size 24 square giving complete instructions and directions for using and reading the tables, measures, and special scales.

3.15 Type III, carpenters', try.

3.15.1 Style 1, try. Type III, style 1 squares shall be similar to Figure 3 and comply with the requirements of Table IV for the length of blade specified (see 6.1).

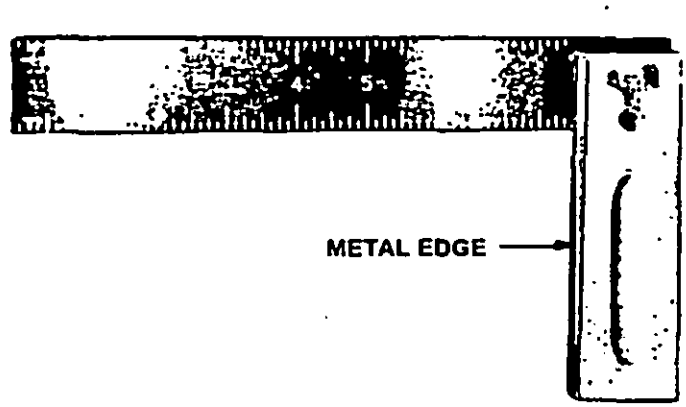


Figure 3. Type III, style 1, carpenters', try.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE IV. Type III, style 1, carpenters' try

Blade				
	Width		Thickness	
Length (Inside of beam)	Min. (Inches)	Max. (Inches)	Min. (Inch)	Max. (Inch)
6	1-5/16	1-13/16	0.042	0.062
8	1-5/16	1-13/16	0.042	0.062
10	1-5/16	1-7/8	0.042	0.062
12	1-5/16	2	0.042	0.062

Beam					
Length		Width		Thickness	
Min. (Inches)	Max. (Inches)	Min. (Inches)	Max. (Inches)	Min. (Inch)	Max. (Inch)
4	5	1-5/16	1-9/16	1/2	5/8
4-7/8	5-5/8	1-5/16	1-9/16	9/16	11/16
5-7/8	7	1-5/16	1-11/16	9/16	11/16
5-7/8	7	1-5/16	1-7/8	9/16	11/16

3.15.2 Style 2, try and miter. Type III, style 2 squares shall be similar to figure 4 and comply with the requirements of table V for the length of blade required (see 6.1).

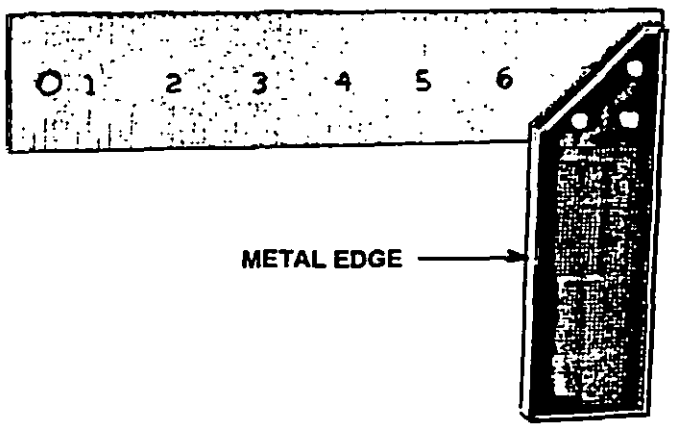


Figure 4. Type III, style 2, carpenters', try and miter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE V. Type III, style 2, carpenters' try and miter

Blade				
	Width		Thickness	
Length (Inside of beam)	Min. (Inches)	Max. (Inches)	Min. (Inch)	Max. (Inch)
6	1-7/16	1-13/16	0.042	0.062
8	1-7/16	1-13/16	0.042	0.062
10	1-7/16	1-7/8	0.042	0.062
12	1-7/16	2	0.042	0.062

Beam					
Length		Width		Thickness	
Min. (Inches)	Max. (Inches)	Min. (Inches)	Max. (Inches)	Min. (Inch)	Max. (Inch)
4	5	1-7/16	1-9/16	1/2	5/8
4-7/8	5-5/8	1-7/16	1-9/16	9/16	11/16
5-7/8	7	1-9/16	1-11/16	9/16	11/16
5-7/8	7	1-9/16	1-7/8	9/16	11/16

3.15.3 Construction. The type III, styles 1 and 2 squares shall consist of a beam of sound and thoroughly seasoned birch, beech, maple, rosewood, cocobolo, goncalo, or similar hardwood and a steel blade of uniform thickness, permanently and securely joined with screws or rivets in a manner such that the alignment will not be changed when subjected to normal use. The blade shall have a minimum hardness of 90 on the Rockwell B scale. The blade shall be made of stainless steel, or shall have a rust-preventative coating on either a blued, black oxide, or natural finish, or shall have an electroplated zinc coating. The beam shall have a smooth finish and shall be bound on the inside surface with a permanently and securely fastened strip of either sheet brass or steel. If a steel binding strip is employed on the beam, it shall be either electroplated with zinc or provided with a chemically- produced oxide coating, oil treated, suitable for resisting corrosion, except that the edges of the strip (not the functional measuring face) may be coated with a clear lacquer or varnish after the surfaces of the beam are finished smooth. Edges of the blade and longitudinal end surfaces of the beam perpendicular to the blade shall be straight and true.

3.15.4 Graduations and figures. Graduations and figures of the type III, styles 1 and 2 squares shall comply with requirements of 3.14.4 for width and depth. Both faces of the blade shall be subdivided to 8ths or 16ths of an inch on the outside edge. Each inch graduation of both faces shall be marked to show the number of inches. Figures on one face shall read from the beam end of the blade; figures on the opposite face shall read either from the beam end of the blade or from the free end of the blade, at the option of the contractor.

3.15.5 Squareness accuracy. The out-of-square error of either edge of the blade with the utile longitudinal end surface of the beam shall not exceed 0.001 inch per inch of length.

3.15.6 Scale accuracy. The accumulative error of graduations shall not exceed 1/32 inch per 12 inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

3.15.7 Finish. The surface finish shall conform to requirements of 3.5.

3.15.8 Miter accuracy. The out-of-miter error of the head with either edge of the blade shall not exceed 15 minutes.

3.16 Type IV, diemakers', with interchangeable blades. Type IV squares shall be similar to Figure 5 and shall conform to the dimensions of Table VI.

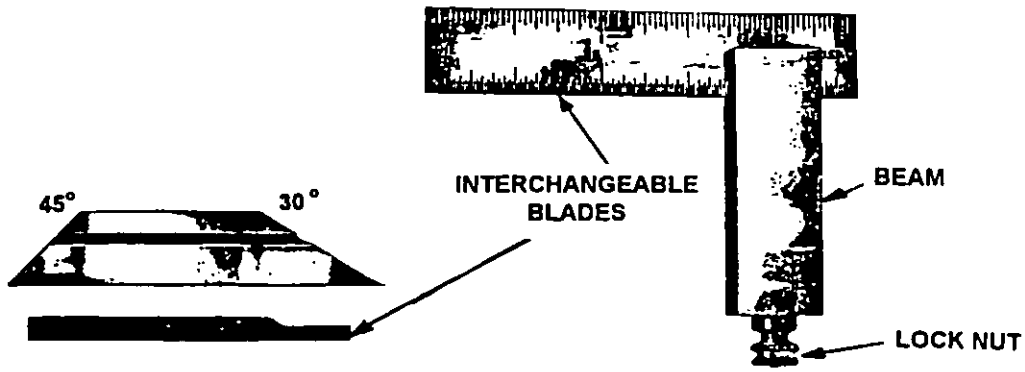


Figure 5. Type IV, diemakers', with interchangeable blades.

TABLE VI. Type IV, diemakers', with interchangeable blades

	Length		Width		Thickness	
	Minimum (Inches)	Maximum (Inches)	Minimum (Inch)	Maximum (Inch)	Minimum (Inch)	Maximum (Inch)
Beam	1-15/32	1/34	25/64	17/32	17/64	19/64
Graduated blade	2-1/2	2-3/4	0.47	0.52	0.055	0.070
Bevel blade	2-1/2	2-3/4	0.47	0.52	0.055	0.070
Narrow blade	2-1/2	2-3/4	0.11	0.16	0.055	0.070

3.16.1 Components. The square shall consist of a graduated blade, bevel blade, narrow blade, and a beam with a blade-clamping device. Blades shall be readily interchangeable and capable of being clamped on the beam.

3.16.2 Blades. Each blade shall be steel, of uniform thickness, properly hardened, tempered, and bright ground. The graduated blade and bevel blade shall be provided with a full length groove along the center of one face for engagement with the clamping device on the beam. Edges and faces shall be straight and true.

3.16.2.1 Graduated blade. The blade shall be graduated on one face only. The upper edge shall be subdivided to 32nds of an inch, and the lower edge to 64ths of an inch. The graduations shall be etched or machine cut to a uniform width and depth. The width shall be from 0.004 to 0.009 inch and the depth shall be from 0.002 to 0.005 inch. Inch graduations shall be numbered. Corners of each graduated edge shall be marked to show the smallest fractional subdivision to which it is graduated. The groove shall be on the ungraduated face.

3.16.2.2 Bevel blade. One end of the blade shall be ground to a 30° angle and the other end to a 45° angle. The blade shall be without graduations.

3.16.2.3 Narrow blade. The narrow blade shall be without graduations.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:41Z
Check the source to verify that this is the currentversion before use.

3.16.3 Beam. The beam shall be steel, hardened, bright ground on all sides, and provided with a slot for insertion of any blade. All surfaces of the beam shall be straight and true and square. The clamping device shall have a substantial knurled thumbscrew or a thumbnut suitable for rigidly clamping the beam at any point on any blade.

3.16.4 Accuracy.

3.16.4.1 Scale accuracy. The scale error taken at the half-inch graduation from each end shall not exceed ± 0.002 inch and the overall scale error shall not exceed + 0.004 inch or - 0.002 inch.

3.16.4.2 Squareness. The out-of-square error of either edge or face of any blade with either longitudinal end surface of the beam shall not exceed 0.001 inch on the edges and 0.002 inch on the faces, measured at the extreme ends of the edges or faces.

3.16.4.3 Parallelism. The out-of-parallel error of opposite longitudinal edges over the entire length of any blade shall not exceed 0.002 inch. The out-of-parallel error of opposite longitudinal end surfaces of the beam which are perpendicular to the blade edges shall not exceed 0.0005 inch over the entire length of the beam.

3.16.4.4 Bevels of bevel blade. The out-of-bevel error for each end of the bevel blade shall not exceed 15 minutes, when assembled in the beam.

3.16.4.5 Finish. The finish of all parts shall conform to the requirements of 3.5.

3.17 Type V, diemakers', with interchangeable blades adjustable at an angle. Type V squares shall be similar to Figure 6 and shall comply with the dimension specified in Table VII.

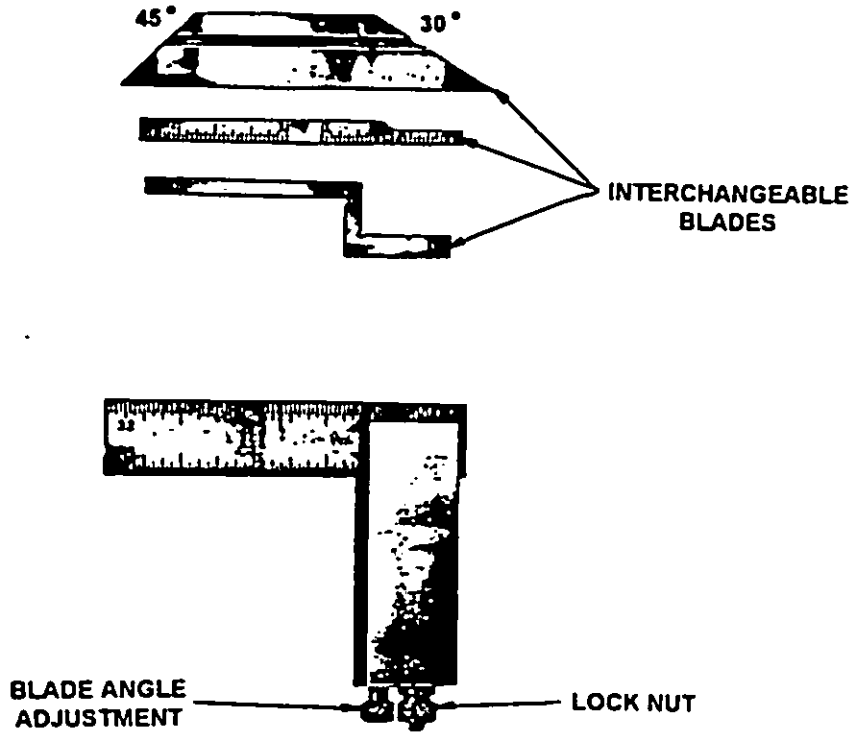


Figure 6. Type V, diemakers', with interchangeable blades adjustable at an angle.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE VII. Diemakers', with interchangeable blades adjustable at an angle

	Length		Width		Thickness	
	Minimum (Inches)	Maximum (Inches)	Minimum (Inch)	Maximum (Inch)	Minimum (Inch)	Maximum (Inch)
Beam	1-11/16	2	9/16	11/16	9/32	11/32
Graduated main blade	2-1/2	2-3/4	0.47	0.52	0.055	0.070
Bevel blade	2-1/2	2-3/4	0.47	0.52	0.055	0.070
Narrow blade	2-1/4	2-3/4	0.14	0.17	0.055	0.070
Offset blade	2	2-3/4	0.11	0.14	0.055	0.070

3.17.1 Components. The square shall consist of a graduated main blade, bevel blade, graduated narrow blade, offset blade, and beam with a clamping device and angularity-adjusting device. Blades shall be readily interchangeable on the beam.

3.17.2 Blades. Blades shall comply with the requirements of 3.16.2.

3.17.2.1 Main graduated blade. The main graduated blade shall comply with the requirements of 3.16.2.1.

3.17.2.2 Bevel blade. The bevel blade shall comply with the requirements of 3.16.2.2.

3.17.2.3 Narrow blade. The narrow blade shall be cut back at one end at a distance of at least 5/8 inch, making the end width 3/32 inch. The lower edge of one face shall be subdivided to 32nds of an inch, and the opposite face may be either with or without 64ths inch graduations, at the option of the contractor. The width and depth of the graduations shall be as specified in 3.16.2.1. Scales may be graduated to read from one end or both ends of the blade, at the option of the contractor. At least one corner of each scale shall be marked to show the smallest subdivision to which it is graduated.

3.17.2.4 Offset blade. Both edges of the offset blade shall be beveled except at the beam shank. When properly inserted in the beam, the offset length shall extend 1-7/16 inches from the beam.

3.17.3 Beam. The beam shall be steel, hardened, bright ground on all sides, and provided with a slot for insertion of any blade. All surfaces of the beam shall be straight and true and square. There shall be two substantial knurled thumbscrews or thumbnuts, one shall be suitable for rigidly clamping any blade at a 90° angle, and the other, when used in conjunction with the clamping screw, shall be suitable for setting and rigidly clamping the narrow blade and the offset blade at any angle up to 8° on either side of the normal square position.

3.17.4 Accuracy. Accuracy shall comply with all requirements as specified in 3.16.4.1 to 3.16.4.4 inclusive.

3.17.5 Finish. The finish of all parts shall conform to the requirements of 3.5.

3.18 Type VI, machinists', combination. Type VI squares shall be similar to Figure 7 and conform to the dimensions specified in Table VIII for length of blade specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:41Z
Check the source to verify that this is the current version before use.

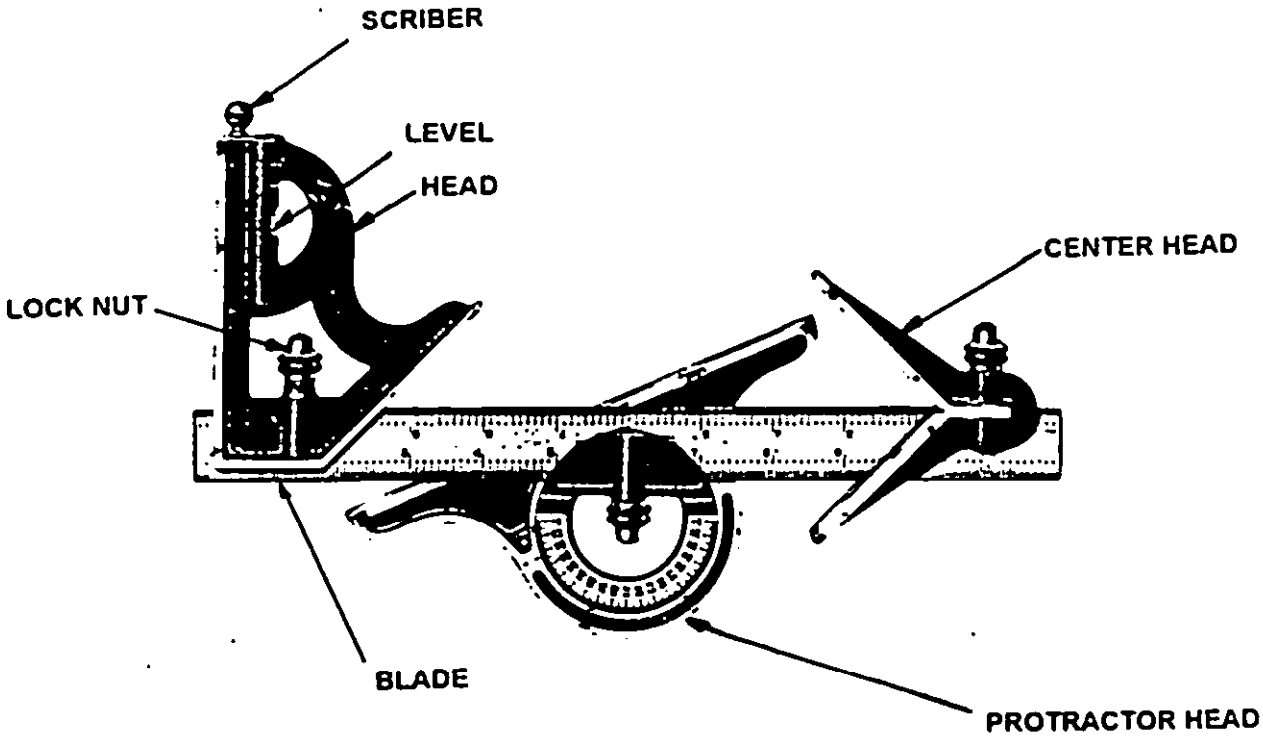


Figure 7. Type VI, machinists' combination (for center head and protractor head, see 3.18.1).

TABLE VIII. Type VI, machinists', combination.

Length	Blade				Square and miter head, length	
	Width		Thickness			
	Minimum (Inch)	Maximum (Inches)	Minimum (Inch)	Maximum (Inch)	Minimum (Inches)	Maximum (inches)
6	11/16	15/16	0.075	0.095	2-7/8	3-1/8
9	13/16	1	0.075	0.095	3-5/8	4-5/16
12	15/16	1-1/8	0.075	0.095	4	4-3/4
18	15/16	1-1/8	0.075	0.095	4-1/4	4-3/4
24	15/16	1-1/8	0.075	0.095	4-1/4	4-3/4

3.18.1 Components. The square shall consist of a graduated blade, square and miter head with a blade-clamping device, level glass, and a scriber. When specified, a center head and protractor head shall also be furnished (see 6.1). (Note: Protractor heads are not available commercially for the 6 inch and 9 inch square.)

3.18.1.1 Class A, square and miter head; cast iron. The square and miter head of class A squares shall be a fine grained, seasoned gray iron casting of uniform quality, free from blowholes, porosity, hard spots, or other detrimental defects. If a center head is called for with class A square, it shall be of cast iron. If a protractor head is

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:41Z
Check the source to verify that this is the current version before use.

called for with the class A square, the body shall be cast iron and of the single (nonreversible) or double (reversible) type, as specified by the procuring agency. If the type of body is not specified, the single (nonreversible) type shall be furnished (see 6.1).

3.18.2 Blade. The blade shall be steel, of uniform thickness with a flatness of 0.010 TIV, properly hardened, tempered, and bright polished. Edges and faces shall be straight and true. A full-length groove along the center of one face shall be provided for engagement with the clamping device of each head.

3.18.3 Blade graduations. Blade graduations shall be style B described as follows, unless style A graduations are specified. The width and depth of graduations shall be in accordance with 3.16.2.1. The subdivisions shall be marked with proper figures at the corresponding inches. Figures on each face shall read in reverse directions. Corners of each graduated edge of each face shall be marked to show the smallest fractional subdivision of which it is graduated. Inch graduations shall be numbered.

3.18.3.1 Style A graduations. One face shall be subdivided to 8ths of an inch on one edge and to 16ths of an inch on the other edge; the other face shall be subdivided to 32nds of an inch on one edge and 64ths of an inch on the other edge.

3.18.3.2 Style B graduations. One face shall be subdivided to 32nds of an inch on one edge and to 64ths of an inch on the other edge; the other face shall be subdivided to 50ths of an inch on one edge and to 100ths of an inch on the other edge.

3.18.4 Square and miter head. The square, miter, and horizontal surfaces, including their adjacent side surfaces, shall be accurately finished. There shall be a slot for insertion of the blade. The clamping device shall have a substantial knurled thrumbscrew or thumbnut suitable for rigidly clamping the head at any point on the blade. The locking device shall be so designed to prevent slipping from the head when the blade has been removed. The level glass shall be parallel with the square face of the head. The scriber shall be hardened steel, pointed, and provided with a knurled head. The scriber shall be either mechanically or frictionally held in a brass bushed hole in the head.

3.18.5 Center head. When specified (see 6.1), a center head shall also be furnished with the square. The inside area surfaces, including their adjacent side surfaces, shall be finished straight and true and square. The arms shall include a 90° angle with a tolerance of $\pm$ 15 minutes, the vertex of which shall center with the edge of the blade. Either arm of the center head shall be at one-half the included angle $\pm$ 3 minutes with either edge of the blade. There shall be a slot for insertion of the blade and a substantial knurled thrumbscrew or thumbnut suitable for securely holding the head at any point on the blade. The locking device shall be so designed to prevent slipping from the head when the blade has been removed. On 6 inch blade squares, the arms of the center head, measured along the inside surface, shall not be less than 1-7/8 inches nor more than 2-1/8 inches long; on 9 inch blade squares, the arms shall be not less than 2-1/8 inches nor more than 2-3/8 inches long; on squares with 12 inch or longer blades, the arms shall be not less than 2-1/4 inches and not more than 2-5/8 inches long.

3.18.6 Protractor head. When specified (see 6.1), a protractor head shall also be furnished with the square. When used in conjunction with the graduated blade, it shall be suitable for use as a bevel protractor. The protractor head shall consist of a body, a turret-type dial with level glass, and two or more clamping devices with substantial knurled thumbscrews or thumbnuts. The locking device shall be so designed to prevent slipping from the head when the blade has been removed. One clamping device shall be suitable for rigidly clamping the protractor body to the blade, the other shall be suitable for rigidly clamping the protractor body to the dial.

3.18.6.1 Body. The straight edges and finished portion of sides of the protractor body shall be finished straight and true and square and shall be not less than 6-3/8 inches nor more than 7-1/4 inches long. The body shall have one or more zero index lines, each marked with the figure zero, for use as reference points with the dial. The single (nonreversible) type of body shall have a single straight edge arranged to extend over one face of the graduated blade and capable of complete rotation about the dial. The double (reversible) type of body shall have two equal length straight edges, each extending equal distances over opposite faces of the graduated blade. One straight edge of the double (reversible) type body may be detachable, at the option of the contractor.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the currentversion before use.

3.18.6.2 Dial. The dial shall be cast iron, bright ground, chrome-plated satin finish, or a stainless steel plate which is permanently secured to the cast iron dial and accurately fitted inside a circular space in the body. The peripheral surface of the dial and the inside surface of the circular space of the body shall be accurately finished. The dial shall be graduated over an arc of 180°, and graduations and figures shall be arranged to read from zero to 90° in either direction at the zero index, or from zero to 180° with two sets of figures reading in opposite directions. Angle graduations shall be subdivided to 1° and marked to show each 10°. The spirit level shall be permanently secured to the dial and arranged parallel to the straight edges of the blade. Graduations shall have durable filling of black color.

3.18.7 Accuracy.

3.18.7.1 Blades and scales. The zero end of the blade end is defined as that blade end nearest the lowest graduation. The leading edge of a graduation is defined as that graduation edge closest to the blade end being used in the measurement.

3.18.7.1.1 Squareness of ends. Both blade ends shall be square to both blade edges within an angle of 90° ± 5 minutes.

3.18.7.1.2 Blade length tolerance. The blade length tolerance for 6 inch blades shall be + 0.004 to - 0.002 inches; for 9, 12, and 18 inch blades it shall be + 0.005 to - 0.020 inches; and for 24 inch blades, it shall be + 0.007 to - 0.012 inches.

3.18.7.1.3 Scale centering error. The scale centering error shall be measured from the leading edge of the first 1/2 inch graduation to the zero blade end and from the leading edge of the last 1/2 inch graduation to the opposite blade end. The scale centering error shall not exceed ± 0.0025 inches for each measurement.

3.18.7.1.4 Scale graduation error. Throughout the first 6 inches, the scale graduation error measured between the left scale end and any graduation shall not exceed + 0.004 to - 0.0035 inches. Throughout greater than 6 inches to 12 inches inclusive, the scale graduation error measured between the left scale end and any graduation shall not exceed + 0.005 to - 0.0035 inches. Throughout greater than 12 inches to 18 inches inclusive, the scale graduation error measured between the zero scale end and any graduation shall not exceed + 0.006 to - 0.0035 inches. Throughout greater than 18 inches to 24 inches inclusive, the scale graduation error measured between the zero scale end and any graduation shall not exceed + 0.007 to - 0.0035 inches.

3.18.7.2 Miter. The out-of-miter error of the head with either edge of the blade shall not exceed 0.004 inch per foot for class A heads.

3.18.7.3 Protractor. The error in any angular measurement on the scale with either graduated blade edge and either straight edge of the protractor body shall not exceed 15 minutes.

3.18.7.4 Center head. The blade shall bisect the included angle, as measured, of the center head within a tolerance of ± 3 minutes.

3.18.8 Finish. The finish of all parts shall conform to the requirements in 3.5, except that utile surfaces of the center head shall have a finish of 100 microinches roughness height rating or better.

3.19 Type VII, machinists' double square, with interchangeable blades. Type VII squares shall be similar to Figure 8 and shall conform to Table IX for the length of graduated blade specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

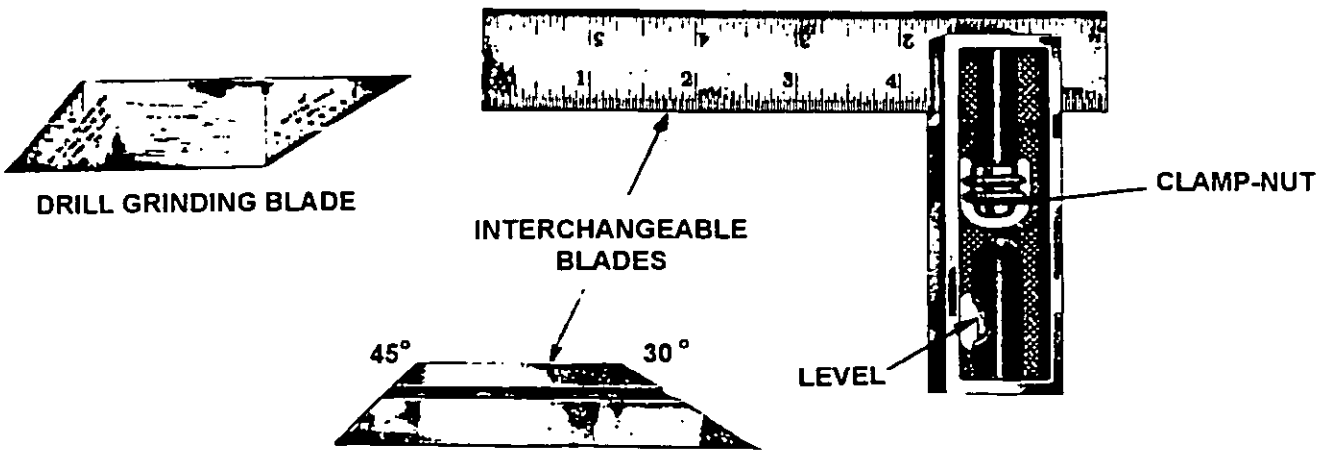


Figure 8. Type VII, machinists', double square, with interchangeable blades.

TABLE IX. type VII, machinists', double square, with interchangeable blades

Length of graduated blade (Inches)	Length of bevel blade		Width of blades		Thickness of blades		Length of head	
	Min (Inches)	Max (Inches)	Min (Inch)	Max (Inch)	Min (Inch)	Max (Inch)	Min (Inches)	Max (Inches)
4	3-3/8	3-3/4	9/16	11/16	0.060	0.095	2-1/8	2-1/2
6	4-5/8	5	13/16	1	0.075	0.095	3-3/8	3-3/4

3.19.1 Components. The square shall consist of a graduated blade, bevel blade, and head with a blade-clamping device. On the 6 inch blade size, the head shall be provided with a permanently secured level glass arranged parallel to the longitudinal end surfaces of the head. When specified (see 6.1), a drill grinding blade shall also be furnished with a 6 inch square.

3.19.2 Graduated blade. The blades shall be steel, of uniform thickness, properly hardened, tempered, and bright polished. Edges and faces shall be straight and true. A full length groove along the center of one face shall be provided for engagement with the clamping device on the head. One face shall be subdivided to 8ths of an inch on one edge and to 16ths of an inch on the opposite edge; the other face shall be subdivided to 32nds of an inch on one edge and to 64ths of an inch on the opposite edge. Width and depth of graduations shall be in accordance with 3.16.2.1. Figures on each face shall read in reverse directions. Corners of each graduated edge on each face shall be marked to show the smallest fractional subdivision to which it is graduated. Inch graduations shall be numbered.

3.19.3 Bevel blade. The blade shall be steel, of uniform thickness, properly hardened, tempered, and bright polished. A full length groove along the center of one face shall be provided for engagement with the clamping device on the head. The blade shall be without graduation. Edges and faces shall be furnished straight and true. One end shall be beveled to a 60° angle and the other end shall be beveled to a 45° angle.

3.19.4 Drill grinding blade. When specified (see 6.1), a drill grinding blade shall also be furnished. When used in conjunction with the head of the square, it shall form a drill grinding gauge. The blade shall be of steel, of uniform thickness, hardened, tempered, and bright polished. It shall be not less than 4-1/2 inches overall in length, and not less than 0.075 inch in thickness. A full length groove along the center of one face shall be provided for engagement with the clamping device on the head. The edges and faces shall be finished straight and true. One end shall be beveled to an angle of 59° and the other end shall be beveled to an angle of 41°. Each beveled end shall have graduations on the ungrooved face of the blade representing distances to 64ths of an inch from the longitudinal

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

end surface of the clamp-on head that is to measure distances perpendicular to the axis of any cylindrical member placed against the longitudinal end surface of the head. Graduations shall cover a scale length representing at least 1-1/16 inches on the 59° end and 5/8 inch on the 41° end, beginning with a zero index line suitable for alignment with the longitudinal end surface of the head. Each inch graduation shall be marked, the corner of each graduated edge shall be marked to show the smallest fractional subdivision represented, and each graduation line representing an 8th inch shall be marked to read the number of 64ths. The 59° bevel shall be permanently engraved with the angle, and the legend "Cutting Angle of Drill"; the 41° bevel shall be permanently engraved with the angle and legend "Cutting Angle of Countersink".

3.19.5 Head. The head shall be steel or cast iron as specified. All surfaces and edges shall be accurately finished. There shall be a slot for insertion of either blade. The clamping device shall have a substantial thrumbscrew or thumbnut suitable for rigidly clamping the head to any point on either blade. The level glass shall be parallel with the longitudinal end surface.

3.19.6 Accuracy.

3.19.6.1 Scale accuracy. Scale accuracy shall conform to requirements in 3.18.7.1.

3.19.6.2 Squareness. The out-of-square error of either edge of the blade with either longitudinal end surface of the beam shall not exceed 0.0015 inch on the edge and measured at the ends of the edge.

3.19.6.3 Parallelism. The out-of-parallel error of opposite longitudinal edges over the whole length of any blade shall not exceed 0.00025 inch. The out-of-parallel error of opposite longitudinal end surfaces on the beam which are perpendicular to the blade shall not exceed 0.00025 inch over the whole length of the beam.

3.19.6.4 Bevels of bevel blade and drill grinding blade. The out-of-bevel error for each end of the bevel blade and drill grinding blade shall not exceed 15 minutes.

3.19.7 Finish. The finish of all parts shall conform to requirements in 3.5.

3.20 Type VIII, machinists', thin steel. Type VIII squares shall be similar to figure 9 and conform to dimensions specified in table X for the size of square.

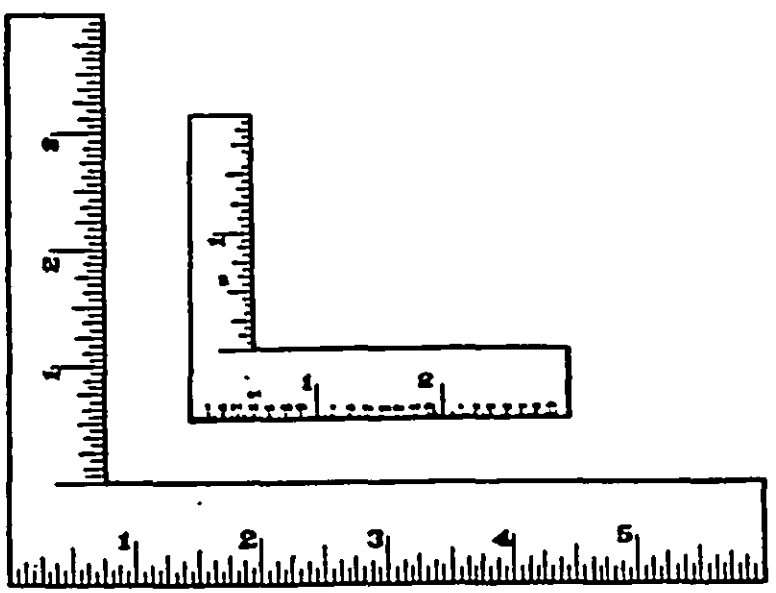


Figure 9. Type VIII, machinists', thin steel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE X. Type VIII, machinists', thin steel

Size (Length of long and short leg)	Width				Thickness of legs 1/	
	Long leg		Short leg			
	Minimum (Inch)	Maximum (Inch)	Minimum (Inch)	Maximum (Inch)	Minimum (Inch)	Maximum (Inch)
2 by 1	7/16	1/2	11/32	13/32	0.048	0.052
3 by 2	19/32	21/32	15/32	21/32	0.048	0.052
4 by 3	11/16	13/16	9/16	13/16	0.051	0.064
6 by 4	13/16	1516	11/16	15/16	0.061	0.064
8 by 6	1-1/16	1-3/16	1-1/16	1-3/16	0.061	0.064

1/ The maximum difference in any thickness measurements made on an individual square shall not exceed 0.005 inch.

3.20.1 Construction. The square shall be steel, in one piece, and bright polished. Edges shall be straight and true; both faces shall be fully flat.

3.20.2 Graduations. Only the inner edge of the short leg and the outside edge of the long leg shall be graduated. On 2 by 1 and 3 by 2 inch squares, the long leg shall be subdivided to 64ths of an inch on one face and 32nds of an inch on the opposite face, and the short leg shall be subdivided to 16ths of an inch on one face and to 64ths of an inch on the opposite face. On squared 4 by 3 inches and larger, the long leg of one face shall be subdivided to 32nds of an inch and the short leg on the same side to 16ths of an inch, and the arrangement of subdivisions on the opposite faces shall be reversed, long leg subdivided to 16ths of an inch and short leg subdivided to 32nds of an inch. Scales subdivided to 64ths of an inch may be with or without markings to read the number of 64ths at each 8th inch subdivision on the scale. The width and depth of graduations shall be in accordance with 3.16.2.1. The subdivisions shall be marked with proper figures at the corresponding inches. Corners of each graduated edge of each face shall be marked to show the smallest fractional subdivision to which it is graduated. Inch graduations shall be numbered.

3.20 3 Accuracy.

3.20.3.1 Scale accuracy. Scale accuracy shall conform to requirements in 3.18.7.1.

3.20.3.2 Squareness accuracy. The out-of-square error of either edge of the long leg with either edge of the short leg shall not exceed the following when measured at the extreme ends of the edges:

	Inch
2 by 1 inch square	0.0010
3 by 2 inch square	0.0015
4 by 3 inch square	0.0015
6 by 4 inch square	0.0020
8 by 6 inch square	0.0025

3.20.4 Finish. The finish shall conform to the requirements in 3.5.

3.21 Type IX, machinists', try. Type IX squares shall be similar to Figure 10 and shall comply with dimensions specified in Table XI for the nominal length of blade specified (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the current version before use.

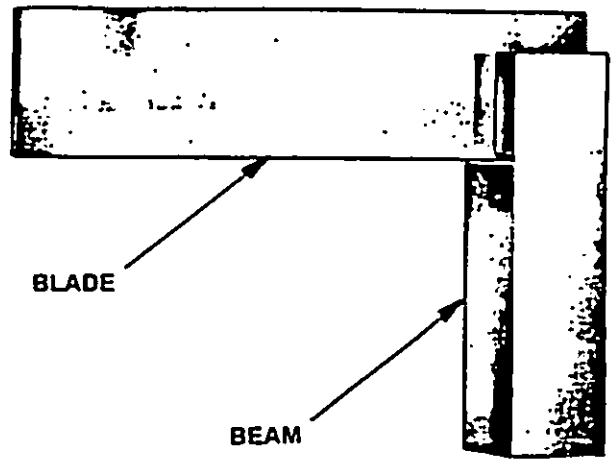


Figure 10. Type IX machinists', try.

TABLE XI. Type IX, machinists', try

Blade				
Length (Inside of beam)		Width	Thickness	
Min. (Inches)	Max. (Inches)	Min. (Inches)	Min. (Inch)	Max. (Inch)
1-1/2	1-15/32	15/32	0.050	0.060
3	2-15/16	21/32	0.060	0.070
4-1/2	4-3/8	1-1/16	0.070	0.100
6	5-7/8	1-5/16	0.070	0.100
9	8-7/8	1-5/8	5/64	3/32
12	11-7/8	1-15/16	5/64	1/8
15	14-7/8	2	3/32	1/8
18	17-7/8	2-13/16	7/64	9/64

Beam		
Length	Width	Thickness
Min. (Inches)	Min. (Inches)	Min. (Inches)
1-1/8	1/2	1/4
1-3/4	23/32	3/8
2-9/16	7/8	7/16
3-1/4	1-1/32	1/2
4-3/8	1-1/4	5/8
5-3/8	1-1/2	3/4
5-11/16	1-7/8	7/8
6-15/16	2-5/16	1-1/8

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:41Z
Check the source to verify that this is the current version before use.

3.21.1 Cases. Instruments shall be furnished in a wood or vinyl covered metal case. The case shall be fitted to hold all parts of the instrument in such a manner as to preclude damage to any parts thereof during ordinary usage and handling and shall be well made. Blocks shall be installed in wooden cases to prevent movement of square in the case. Wooden cases shall have a hinged cover and clasp or a sliding cover and shall be coated both inside and out with a good protective coating of lacquer or varnish which will not absorb, affect, or be affected by the compound specified for preservation of the items. Metal cases shall have a spring loaded piano-type hinge. The case shall be vinyl covered with expanded polyethylene foam heat impressed inserts in base and cover. The impressions shall conform to the configuration of the instrument, thus making a secure case for shipment and storage with ample cushioning.

3.21.2 Construction. The square shall consist of a steel blade and steel beam, both hardened, tempered, ground, lapped and polished, and of uniform thickness. The blade shall be closely fitted and permanently secured to the beam with rivets or dowel pins by sweating or combination thereof, in a manner which as to assure permanent squareness. There shall be no excessive gaps between the blade and beam. including fill-ins with solder. The strength of any other type of construction shall equal or exceed the strength in sheer of a sweat-soldered assembly, and the heads of pins or rivets shall be ground flush or be below the beam surface. The inner corner of the beam, adjacent to the lower blade edge, shall be grooved to afford a clearance of burrs. The groove shall extend across the entire beam width. Edges and faces of the blade, and faces and edges of the beam, shall be finished straight and true. The blade shall be without graduations. Longitudinal blade edges shall be flat or beveled, as specified (see 6.1). When beveled edges are specified, both edges shall be beveled on each side so as to practically make line contact with the work.

3.21.3 Accuracy.

3.21.3.1 Squareness. The out-of-square error of either edge of the blade with either longitudinal end surface of the beam shall not exceed the following when measured at the extreme ends of the edges:

		Inch
1-1/2	inch blade	0.0001
3	inch blade	0.0001
4-1/2	inch blade	0.0001
6	inch blade	0.00015
9	inch blade	0.00015
12	inch blade	0.0002
15	inch blade	0.0002
18	inch blade	0.0002

3.21.3.2 Parallelism. The out-of-parallel error of opposite blade edges and out-of-parallel error of longitudinal end surfaces of beam shall not exceed the following when measured at the extreme ends:

		Inch
1-1/2	inch blade	0.0001
3	inch blade	0.00015
4-1/2	inch blade	0.00015
6	inch blade	0.0002
9	inch blade	0.0002
12	inch blade	0.0003
15	inch blade	0.0003
18	inch blade	0.0003

3.22 Finish. The finish of both blade and beam shall conform to requirements in 3.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

3.23 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the specification preparing activity for change to this document.

3.24 Workmanship: Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth, dry, adherent, continuous, and not stained or discolored. Fasteners shall be firmly secured and show no evidence of deformation, cross threading, or hazardous burrs. Adhesives and lubricants adequate for their intended purpose shall be properly and neatly applied. Adhesives shall be adequately cured. Wires and cables shall be neatly dressed and shall not be frayed or in contact with sharp edges. Wire and cable insulation shall show no evidence of burns, abrading, or pinch marks. There shall be no interference, binding, or galling of parts. External and bearing surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be clean and free of corrosion and debris (e.g., chips, shavings, slivers) or other foreign material. The item shall be free from manufacturing workmanship defects (e.g., loose, missing, binding or misaligned parts, sharp or rough external edges, corners, or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears, nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specification and standards to the extent specified, or, if none, in accordance with this specification.

4.2 Sampling procedures. Unless otherwise specified, sampling for inspection shall be performed in accordance with ANSI/ASQC Z1.4. Inspection levels and acceptance quality levels (AQL) shall be as specified in Table XII.

TABLE XII. Inspection levels and AQLs

Category	Inspection level	AQL	Reference
Visual and dimensional examination (except 4.4.1)	II	Major 1.5 Minor 2.5	4.3
Testing	S-4	1.0	4.4

4.3 Visual and dimensional examination. Each sample square selected in accordance with 4.2 shall be examined for defects listed in Table XIII. The AQL shall be expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:41Z
Check the source to verify that this is the currentversion before use.

TABLE XIII. Classification of defects

Examine	Defect	Classification Critical	Major	Minor
Construction and workmanship	None defined		001	
	Type, length, and width (of blade not as specified		101	
	Type, class, style, and size of square not as specified			102
	Materials defective or not as specified; evidence of rust, fins, burrs, nicks, or cracks			103
	Blades defective, not of uniform thickness, not polished, edges not finished, or blade(s) not as specified (types I, IV, V, VI, and VII only)			104
	Head or beam defective, not finished as required, clamping device defective, clamping nut, screwhead) not knurled, or head or beam not as specified			105
	Head or center head cannot be mounted at either end of the blade(s), head sticks or binds on the blade(s), or head and blade(s) not interchangeable with others of the same manufacturer (types I, VI, and VII only)			106
	Level glass missing, broken, not permanently secured, not enclosed as required, liquid discolored or cloudy, or glass not etched as required (types I, VI, and VII only)			107
	Square not one piece, not flat, or not finished as required (types II and VIII only)			108
	Protractor head (if applicable) defective, not finished as required, clamping devices defective, clamping nuts (or screwheads) not knurled, dial not accurately fitted in the body, spirit level defective or not permanently secured, or protractor head not as specified (type VI only)			109
	Thickness of tongue or body at end less than the specified minimum (type II, classes A and B, style 1 only)			110
	Thickness of body and tongue not uniform throughout (type II, classes A and B, style 2 only)			111
	Difference in thickness exceeds the specified maximum (type VIII only)			112
	Beam not permanently and securely attached to the blade as required, evidence of looseness (types III and IX only)			113
	Beam not bound on the inside edge, or metal strip not securely fastened to the beam (type III only)			114
	Rivets or dowel pins (for securing beam to blade) not ground flush, inner corner of beam not grooved, evidence of excessive gaps between blade and beam, or gaps (between beam and blade) filled with solder (type IX only)		115	
	Beam and blade not finished and coated (if required) as specified			116

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE XIII. Classification of defects (cont'd)

Examine	Defect	Classification Critical	Major	Minor
Construction and workmanship (cont'd)	Blades not interchangeable, blades stick or or bind in the beam slot, or blades and beam not interchangeable with others of the same manufacturer (types IV and V only)		117	
	Adjusting screw (on beam) defective, not knurled, not capable of adjusting the angle between the beam and the narrow or offset blades, or amount of angular adjustment not as required (type V only)		118	
	Narrow blade not cut back (at one end) the required distance (type V only)		119	
	Edges of offset blade not beveled as required or length of offset not as required (type V only)		120	
	Thickness, width, and length of blades not within the specified values		121	
	Graduations or figures illegible, incorrect, not sharply defined, not straight, not uniform; graduations not etched, machine cut or stamped, inch graduations, not numbered or graduations not as specified (except type IX)		122	
	Length of head or beam not within the specified maximum and minimum values			201
	Thickness where tongue and body join exceeds the specified maximum (type II, classes A and B, style 1 only)			202
	Instruction booklet missing or not as specified (type II, 24 inch size only)			203
	Length, thickness and width of beam not within the specified maximum and minimum values (types III, IV, V, and IX only)			204
	Length of arms on the center head (if applicable) not within the specified maximum and minimum values (type VI only)			205
	Length of straight edge(s) on the protractor head (if applicable) not within the specified maximum and minimum values (type VI only)			206
	Marking, country or origin and manufacturer's part number and name or trademark missing, illegible, incorrect, or not permanent			207

4.4 Testing. Each sample unit selected in accordance with 4.2 shall be tested in accordance with 4.4.1 through 4.4.7. The AQL shall be expressed in terms of defects per hundred units.

4.4.1 Dimensions. Each sample square shall be tested at a temperature of 68°F whenever practical be, to determine conformance with the applicable dimensional requirements of this specification. Readings taken at other temperatures shall be adjusted to 68°F.

4.4.2 Out-of-squareness. All sample squares shall be tested to determine their squareness. Squareness shall be determined by means of a surface plate and a precision cylinder square used with a comparator-square or with a master square of known accuracy. Types I, II, and III squares may be tested with a precision square and a surface plate.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the current version before use.

4.4.3 Squareness test for inside. All sample squares shall be tested to determine their accuracy. A surface plate and angle plate may be used with a thickness gauge to measure the error.

4.4.4 Angular trueness. All sample type VI squares with protractor heads shall be tested for angular intervals. This shall be determined by means of either a circular measuring machine, sine bar, precision angle blocks, or other methods of equivalent accuracy.

4.4.5 Miter accuracy. Samples of types I and VI squares shall be tested for miter accuracy. This shall be determined by means of either universal angle gauges, or and approved master miter of known accuracy may be used.

4.4.6 Hardness. Hardness testing shall be performed in accordance with ASTM E 18.

4.4.7 Bevel accuracy. Bevel accuracy of the types IV, V, and VII blades and the miter of the type III, style 2 beam shall be tested for bevel accuracy. This shall be determined by means of a universal protractor of known accuracy reading to 5 minutes.

4.4.8 Accuracy. Type IX square shall be checked for accuracy by the use of gauge blocks and angle gauge blocks, or by other means approved by the Government.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, or Commercial, as specified (see 6.1).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type, class, style, size, and blade length required (see 1.2 and applicable tables).
c. Specify when accuracy requirements are greater than those given and state the accuracy required (see 3.12).
d. Style of graduations, if different from those specified (see 3.14.4.1, 3.14.4.3, and 3.18.3).
e. When framing rafter table is required (see 3.14.6).
f. Attachments required (see 3.18.1.1, 3.18.1.2, 3.18.5, 3.18.6, 3.19.1, and 3.19.4).
g. Whether blades shall be flat or beveled (see 3.21.2).
h. Special finish for graduated blades and protractor heads, if required (see 3.5).
i. Level of preservation, packaging, and packing required (see 5.1).
j. Marking, if other than specified.

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:41Z
Check the source to verify that this is the current version before use.